**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Iteration: WHILE Loops

**Name** _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

## Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. When is a count-controlled loop suitable?

2. Why use pseudocode?

A) Run directly on the CPU
B) Plan steps without exact syntax
C) Replace all program testing
D) Draw the user interface

3. "A diamond is used only for unconditional calculations." Correct this. Explain why.

4. A program works out a bill, then applies a discount. "The order of steps never matters." Correct this. What could go wrong?

## Today's objectives

Write condition-controlled loops. Apply the programming techniques to a new problem.

## Key vocabulary - Read, cover, recall

**WHILE loop**
Checks a condition before each iteration; repeats while it is true.

**Condition**
A comparison such as attempts < 3 controls continuation.

**Update**
Change loop variables so the condition can eventually become false.

**Unknown repetition**
WHILE suits a task where the number of tries is not fixed beforehand.

**Write one key term and its meaning in your own words:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Learn and practise

## New learning - Key ideas

| | |
|---|---|
| **WHILE loop** | Checks a condition before each iteration; repeats while it is true. For example, consider a password attempt counter. |
| **Condition** | A comparison such as attempts < 3 controls continuation. For example, consider a quiz retry. |
| **Update** | Change loop variables so the condition can eventually become false. For example, consider a low stock checker. |
| **Unknown repetition** | WHILE suits a task where the number of tries is not fixed beforehand. For example, consider a temperature alarm. |

## I Do - One worked example

**A password attempt counter is being developed. Explain how while loop and condition are used. [4 marks]**

1. Identify where this particular program needs control or values.

2. WHILE loop [1]; Checks a condition before each iteration; repeats while it is true. [1].

3. Condition [1]; A comparison such as attempts < 3 controls continuation. [1].

4. Give one accurate point and a linked reason for each technique.

## We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A quiz retry is being programmed. Explain why while loop matters. [2 marks]

2. For a low stock checker, explain how condition is used. [2 marks]

3. A temperature alarm needs update and unknown repetition. Explain the roles of both. [4 marks]

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Think and write

## Code First - Predict > Write > Test > Improve

**YOUR TASK: Add one line so the loop reads another answer and can finish.**

```
answer = input("Continue? Y/N")
while answer != "N"
...
endwhile
print("Stopped")
```

BLACK PEN: predict or complete the code; test, then improve.

## You Do - Independent exam questions

Q2. State TWO concepts needed for this programming technique. [2 marks]

Q3. Explain how condition and update can help a correction prompt. [4 marks]

Q4. Explain the roles of while loop, update and unknown repetition when creating a survey response. [6 marks]

## Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Checks a condition before each iteration; repeats while it is true. ☐ T ☐ F
2. Checks a condition before each iteration; repeats while it is true ☐ T ☐ F
3. A comparison such as attempts < 3 controls continuation ☐ T ☐ F

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

## Do Now - Answers

**Q1** When the number of repetitions is known or determined by a fixed count. ☐

**Q2** B - Plan steps without exact syntax ☐

**Q3** A decision or condition with alternative paths. ☐

**Q4** Steps must be clear and in the right order. Check that conditions give the right result for every allowed input. ☐

## Code First - Model

```
answer = input("Continue? Y/N")
while answer != "N"
answer = input("Continue? Y/N")
endwhile
print("Stopped")
```

**Output: Stopped (test input sequence: Y, then N)**

## We Do - Check the model

1. WHILE loop [1]; Checks a condition before each iteration; repeats while it is true. [1]. ☐

2. Condition [1]; A comparison such as attempts < 3 controls continuation. [1]. ☐

3. Update [1]; Change loop variables so the condition can eventually become false. [1]. Unknown repetition [1]; WHILE suits a task where the number of tries is not fixed beforehand. [1]. ☐

## Exit Ticket - Check

**1. TRUE 2. TRUE 3. TRUE**

## You Do - Self-mark

**Q2** WHILE loop [1]; Update [1]. ☐

**Q3** Condition [1]; A comparison such as attempts < 3 controls continuation. [1]. Update [1]; Change loop variables so the condition can eventually become false. [1]. ☐

**Q4** WHILE loop [1]; Checks a condition before each iteration; repeats while it is true. [1]. Update [1]; Change loop variables so the condition can eventually become false. [1]. Unknown repetition [1]; WHILE suits a task where the number of tries is not fixed beforehand. [1]. ☐

## Try the question again

**1. Copy the FULL question you found most difficult:**

**2. In GREEN PEN, write a complete new answer:**